

claude-windows / 42211f80-2169-4025-ad9d-99c1aa29ff70

Metadata

```
- Source: `claude-windows`  
- Kind: `claude`  
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-inde  
xer\42211f80-2169-4025-ad9d-99c1aa29ff70\subagents\workflows\wf_6237b110-d58\agent-  
a2dc2ab7b8e7d13cd.jsonl`  
- SHA-256: `f9ea74cdb3d53d5085b8569ff8c65e34196ec670b0d6f8d029f7f6db8cd5e524`  
- Source modified: `2026-06-06T17:32:19+00:00`  
- Imported at: `2026-07-05T16:48:28+00:00`  
- project: `wf_6237b110-d58`  
- session_id: `42211f80-2169-4025-ad9d-99c1aa29ff70`
```

Transcript

1. user (2026-06-06T17:31:28.123Z)

Source Extractor

Research question: "Research alternative methods to automatically detect rally start/end segments (temporal segmentation) in racket-sport video ? especially badminton ? for a SELF-HOSTED, LOCAL-FIRST pipeline, and assess whether a cloud-VLM teacher ? local-model distillation is the best approach or whether other methods are competitive or complementary. Keep it focused ("mini" research): prioritize actionable, commercially-licensable, locally-runnable approaches over exotic SOTA needing huge compute or restrictive licenses.

OUR CONTEXT / CONSTRAINTS (ground every recommendation in these):

- Self-hosted on a single consumer GPU (RTX 2070 Super), Windows + WSL2. We want LOCAL-ONLY inference at runtime ? privacy lever: user video never leaves the device. It's a commercial product, so PERMISSIVE / commercial-friendly licenses are REQUIRED (avoid AGPL and non-commercial-only weights/datasets).
- Current substrate: WASB (MIT) shuttle-trajectory detector ? per-frame (x, y, visibility). We derive rally windows via velocity/merge-gap/min-duration thresholds plus a net-crossing gate. We ALSO have a permissive person detector available (torchvision Faster R-CNN + supervision ByteTrack) and the video's AUDIO track.
- Current best approach: use paid Gemini (cloud VLM, full-context) OFFLINE as a TEACHER to label rallies, then DISTILL into a local model (logistic regression on shuttle-trajectory features). Findings so far: threshold-tuning ceilings around F1 ? 0.44 and overfits with few videos; the learned model is data-starved at n=2 labeled videos; WASB candidate RECALL is a real bottleneck (its windows sometimes don't align with true rallies at IoU 0.5).
- Inputs we can exploit locally: video frames, the shuttle trajectory, player bounding boxes / pose (if we add a permissive pose model), and the AUDIO track (shuttle "thwack" hits, crowd noise, umpire/score calls).

SPECIFICALLY COVER:

1. Method families for rally/point segmentation in racket sports (badminton, tennis, table tennis, pickleball): audio-based shot/hit detection; temporal action localization (TAL) deep models; pose/player-stance + court-homography state machines (e.g., "serve-ready stance" gating); classical change-point detection / HMM on the shuttle trajectory; scoreboard / score-overlay OCR for point boundaries; self-supervised / weakly-supervised approaches; and VLM teacher?student distillation.
2. For EACH family: how it works, the labels/data it needs, expected accuracy, local-inference COMPUTE cost, and LICENSE / commercial viability of any models, datasets, or weights involved.
3. Badminton rally-segmentation datasets (e.g., ShuttleSet, TrackNet/TrackNetV2 data) and their licenses / commercial usability.
4. A RANKED recommendation for our local-first, permissive, single-GPU setup: is Gemini-teacher ? local-distillation the best, or are some alternatives competitive or complementary ? especially AUDIO-based hit detection and pose/court-geometry state machines, which we can run fully locally?
5. Concrete, cheap next experiments we could run with what we already have (shuttle trajectory + person detector + audio)."

Fetch and extract key claims from this source:

URL: <https://github.com/qaz812345/TrackNetV3>

Title: qaz812345/TrackNetV3 ? code MIT-licensed, ships pretrained weights, but depends on the non-permissive Shuttlecock Trajectory Dataset

Found via: datasets/licensing search

Task

1. Use WebFetch to retrieve the page content.
2. Assess source quality: primary research/institution? secondary reporting? blog/opinion? forum? unreliable?
3. Extract 2-5 FALSIFIABLE claims that bear on the research question. Each claim must:
 - be a concrete, checkable statement (not vague generalities)
 - include a direct quote from the source as support
 - be rated central/supporting/tangential to the research question
4. Note publish date if available.

If the fetch fails or the page is irrelevant/paywalled, return claims: [] and sourceQuality: "unreliable".

Structured output only.

2. assistant (2026-06-06T17:31:30.672Z)

I'll fetch the TrackNetV3 GitHub page to extract relevant claims.

3. user (2026-06-06T17:31:47.889Z)

TrackNetV3 Repository Analysis

(1) CODE LICENSE

MIT License - The repository explicitly includes an MIT license file.

(2) PRETRAINED WEIGHTS/CHECKPOINTS

- **Availability:** Checkpoints available via Google Drive link (requires download)
- **Files:** `TrackNet\_best.pt` and `InpaintNet\_best.pt`
- **License:** Not explicitly stated in the documentation; inherits from code license (MIT)
- **Access:** "Download the checkpoints" from provided Google Drive link; unzip to `ckpts` directory

(3) DATASET

- Primary Dataset:** Shuttlecock Trajectory Dataset
- **Citation:** Referenced as "Shuttlecock Trajectory Dataset" with link provided
 - **License/Terms:** Not specified in this README; external dataset with separate terms
 - **Structure:** Contains professional and amateur badminton matches with video frames and CSV coordinate labels
 - **Attributes:** Each CSV contains "Frame, Visibility, X, Y"

(4) MODEL FUNCTIONALITY

"A model composed of two core modules: trajectory prediction and rectification" for shuttlecock tracking. Outputs per-frame coordinates (X, Y) and visibility status. Performance: "97.51% Accuracy, 98.56% F1 score" on test set.

(5) HARDWARE/COMPUTE REQUIREMENTS

- **Environment:** Ubuntu 16.04.7 LTS, Python 3.8.7, PyTorch 1.10.0
- **Performance:** "25.11 FPS" on test hardware (unspecified GPU/CPU details)
- **Memory:** Large video support via `--large\_video` flag to prevent memory errors

(6) COMMERCIAL USE RESTRICTIONS

None explicitly stated - MIT License permits commercial use with attribution and liability disclaimer.

4. assistant (2026-06-06T17:31:50.988Z)

Let me verify the dataset license terms by checking the Shuttlecock Trajectory Dataset reference more directly, since the README didn't specify it.

5. user (2026-06-06T17:31:58.873Z)

Dataset and License Information

Shuttlecock Trajectory Dataset URL

The README provides this exact link:

```
**https://hackmd.io/Nf8Rh1NrSrQNUzm00sQKZw**
```

This appears to be hosted on HackMD (a collaborative markdown platform), not a traditional dataset repository like Kaggle or a university server.

Dataset Mention

The exact sentence referencing the dataset:

```
> "Performance on the test split of Shuttlecock Trajectory Dataset."
```

There are also references in the Installation and Training sections directing users to "Download Shuttlecock Trajectory Dataset" at that same HackMD link.

****No license, terms-of-use, non-commercial, or academic-only restrictions are mentioned**** in the provided README content for the dataset itself.

MIT License

The repository lists "MIT license" in its footer navigation, but the exact full license text is not included in the provided content.

Paper Citation

The README references: "[[paper](https://dl.acm.org/doi/10.1145/3595916.3626370)]"

The exact title appears in the page header:

```
> "TrackNetV3: Enhancing ShuttleCock Tracking with Augmentations and Trajectory Rectification"
```

6. assistant (2026-06-06T17:32:02.915Z)

I have enough to assess this source. The README confirms the MIT code license and that the model depends on the Shuttlecock Trajectory Dataset (hosted on HackMD, associated with the AI CUP competition). The README itself does not state the dataset's terms ? so I'll be careful not to overclaim the dataset's non-permissive status from this source alone, and rate that claim as supporting. This is a primary source (the project's own repository).

7. user (2026-06-06T17:32:12.086Z)

Structured output provided successfully

8. assistant (2026-06-06T17:32:19.621Z)

I've extracted and returned the key claims from the TrackNetV3 repository via StructuredOutput. Summary of findings:

- Source quality: primary (the project's own GitHub repository, backed by a peer-reviewed ACM MM Asia 2023 paper).
- The code is MIT-licensed and pretrained weights are downloadable (inheriting MIT by default since no separate weights license is stated) ? relevant as a permissive, locally-runnable alternative/complement to WASB for the shuttle-trajectory substrate.

- Key caveat for the research question: the repo depends on the external Shuttlecock Trajectory Dataset (HackMD/AI CUP), whose terms are NOT restated in the README ? so the source confirms permissive **code/weights** but does NOT establish commercial **dataset** rights. I rated the dataset-dependency claim as central but kept the wording within what this source actually supports (the README is silent on dataset terms rather than affirmatively stating they're non-permissive).
- Output substrate (per-frame Frame/Visibility/X/Y) matches our existing pipeline, runs ~25 FPS on a single GPU with a large-video flag ? feasible on the RTX 2070 Super.